

Aryan Saraf

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EDUCATION

University of Southern California – M.S. Computer Science Expected Dec 2026
Coursework: Networked Systems for Cloud Computing, Analysis of Algorithms, Applied NLP, Multimedia Systems Design
Shah and Anchor Kutchhi Engineering College – B.E. Information Technology May 2023

SKILLS

Languages and Frameworks: Go, Python, Java, C#, Bash, SQL, .NET, Unity, PyTorch
Networking and Systems: TCP/IP, ECN, DCTCP, SDN (OpenFlow), BGP, OSPF, DNS, Linux tc, netlink, Mininet, Open vSwitch
Cloud and Infrastructure: GCP, Kubernetes, Docker, Terraform, Kafka, Azure DevOps, GitHub Actions, ELK
Databases: PostgreSQL, MySQL, Cassandra, ScyllaDB, MongoDB, Redis, Elasticsearch

EXPERIENCE

University of Southern California – Los Angeles, CA
Course Producer, CSCI 526 Advanced Game Consoles Aug 2026 – Present

- Delivered technical lectures on Unity and game systems to 80+ graduate students
- Managed logistics for 17 project teams by coordinating milestone schedules, submission pipelines, and grading, keeping project work organized throughout the course

Course Grader, CSCI 526 Advanced Game Consoles Jan 2026 – May 2026

- Evaluated Unity and C# assignments for 60+ graduate students and gave rubric-based written feedback

Graduate Research Assistant May 2025 – Aug 2025

- Held API latency under 50 ms (local) syncing a C# .NET backend with Unity for an American football simulation game
- Reduced integration time ~40% with reusable Unity prefabs; built facility upgrade flows on SQLite and JSON

Jio Platforms Limited – Mumbai, India
DevOps Engineer Oct 2023 – Dec 2024

- Architected a real-time change-data-capture (CDC) pipeline streaming NoSQL datastore changes into Elasticsearch to power live analytics for enterprise clients
- Reduced deployment time ~80% by automating GCP provisioning with Terraform across 4 standardized environments
- Cut platform latency 30% with Azure DevOps CI/CD pipelines and zero-downtime Kubernetes rollouts
- Saved \$7,000+ monthly by decommissioning idle infrastructure; ran vulnerability and malware scans on 100+ servers
- Operated and tuned 7 database engines for high availability across dev, QA, UAT, pre-prod and production, including Cassandra, ScyllaDB, MongoDB and Redis

PROJECTS

Fabric Microburst Telemetry | github.com/aryansaraf018/fabric-telemetry Go, Python, Mininet, Linux tc, netlink

- Detected 100% of 8 ms congestion bursts with 1 ms queue polling vs. 1% at 1 s, where byte counters reported 17% peak utilization on a link saturated during every burst
- Built a zero-allocation Go agent sampling switch-queue depth over raw netlink every 100 μ s, plus a synchronized incast generator emulating AI collective traffic on a leaf-spine fabric
- Found ECN marking cut median peak queue 29% but left median collective completion unchanged and p99 15% worse; diagnosed a 10.9 ms shaper stall caused by GSO super-packets

Latency-Aware Load Balancer | github.com/aryansaraf018/CS599-Latency-aware-load-balancing Go, Mininet, OpenFlow

- Cut p99 tail latency $3.8\times$ (570 ms to 150 ms vs. Least Connections) under straggler injection with a Go load balancer that routes by real-time active probing (Prequal) on a two-level fat-tree
- Sent up to 39% fewer probes at equal p99 with a controller that adapts probe rate to server heterogeneity
- Isolated network delay as the driver of the gain: on a flat topology, all 4 routing policies tied at ~43 ms p99